Yuliang(Leo) Chen

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Education

Dartmouth College, Hanover — Hanover, New Hampshire
Ph.D. Student in Computer Science — *Sep 2025 – Jun 2030*

University of California, San Diego — San Diego, California
M.S. in Data Science — *Sep 2023 – Jun 2025*

University of California, Santa Barbara — Santa Barbara, California
B.S. in Statistics – Data Science | B.S. in Mathematics — *Sep 2018 – Jun 2022*

Research Interests: Multimodal representation learning and reasoning, self-supervised learning, AI4Health (Bio Sensing)

Publications

Lapras: Latent Reasoning for Time Series Language Models
Yuliang Chen, Yu Yvonne Wu, Patrick Langer, Arvind Pillai, Sudarshan Regmi, Martin Maritsch, Juncheng Liu, Robert Jakob, Thomas Kaar, Tess Z. Griffin, Lisa Marsch, Michael V. Heinz, Nicholas C. Jacobson, and Andrew Campbell. *Under review*, 2026.

Learning Transferable Sensor Models via Language-Informed Pretraining
Yuliang Chen, Arvind Pillai, Yu Yvonne Wu, Tess Z. Griffin, Lisa Marsch, Michael V. Heinz, Nicholas C. Jacobson, and Andrew Campbell. *arXiv preprint arXiv:2603.11950*, 2026.

Toward Foundation Model for Multivariate Wearable Sensing of Physiological Signals
Yunfei Luo*, **Yuliang Chen***, Asif Salekin, and Tauhidur Rahman. *ACM Transactions on Computing for Healthcare (HEALTH)*, 2025.

Multi-modal Cross-masked Autoencoder for Digital Health Measurements
Howon Ryu*, **Yuliang Chen***, Yacun Wang, Andrea Z. LaCroix, Loki Natarajan, Yu Wang, and Jingjing Zou. *arXiv preprint arXiv:2506.02260*, 2025.

BALMS: Benchmarking Agentic LLMs for Longitudinal Mental Health Sensing
Yu Yvonne Wu, Arvind Pillai, **Yuliang Chen**, Yuwei Zhang, Sudarshan Regmi, Tess Z. Griffin, Michael V. Heinz, Lisa A. Marsch, Nicholas C. Jacobson, and Andrew Campbell. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

Toward World Modeling of Physiological Signals with Chaos-Theoretic Balancing and Latent Dynamics
Yunfei Luo, Xi Chen, **Yuliang Chen**, Lanshuang Zhang, Md Mofijul Islam, Siwei Zhao, Peter Kotanko, Subhasis Dasgupta, Andrew Campbell, Rakesh Malhotra, and Tauhidur Rahman. *arXiv preprint arXiv:2605.15465*, 2026.

* *Equal contribution; order decided by a coin flip.*

Research Experience

HealthX Lab, Dartmouth College — Sep 2025 – Present
Graduate Research Assistant; Advisor: Prof. Andrew Campbell — *Hanover, NH*

- Developed **Lapras**, a post-training framework that enables LLMs (e.g., Qwen3-8B) to reason over time series in latent representation space rather than explicit text, improving average F1 by **up to 10.8%** over standard chain-of-thought fine-tuning while generating **23.9× fewer tokens**.
- Pretrained **SLIP**, a **unified sensor–language model** that learns a shared sensor–text embedding space from **heterogeneous** sensor data on **600K sensor–caption pairs**, surpassing the state-of-the-art time series foundation model by **5.7%** in linear-probe accuracy across **11 datasets**.
- **Instruction-tuned** SLIP for open-ended sensor question answering and captioning, surpassing the state-of-the-art time series language model by **7.5%** in average QA accuracy across 4 benchmarks.

- Co-developed **BALMS**, the first **benchmark of LLM agents** for wellbeing prediction from longitudinal wearable data, comparing prompt-based, tool-augmented, and RAG agents across 5 LLMs and 3 real-world datasets.

Zou Lab, UC San Diego Jul 2024 – Sep 2025

Graduate Student Researcher; Advisor: Prof. Jingjing Zou *San Diego, CA*

- Pretrained **MoCA**, a theoretically grounded **multimodal masked autoencoder** for longitudinal wearable data, improving linear-probe accuracy by **15.0%** over prior methods across 6 transfer benchmarks.
- Shipped an open-source **lightweight sitting-detection model** for wrist sensors, building an end-to-end pipeline over **8.2K hours** of real-world data and improving balanced accuracy by **10.7%** over the prior published model.

MoSAIC Lab, UC San Diego Mar 2024 – Jun 2025

Research Assistant; Advisor: Prof. Tauhidur Rahman *San Diego, CA*

- Pretrained **NormWear**, a **sensor-agnostic foundation model** for multivariate wearable signals, outperforming existing time series foundation models on **18 tasks across 11 datasets**.
- Co-developed **NormWear-2**, a **world model** that captures physiological signals and clinical interventions through shared latent dynamics, outperforming state-of-the-art time series foundation models by **up to 13.5%** in composite forecasting score across 5 datasets spanning millisecond to hourly resolutions.

Skills

Languages: Python, SQL, R, C/C++ **Infra:** Slurm, Ray, Kubernetes, Docker, Hydra, Weights & Biases

Training: PyTorch (DDP, FSDP), DeepSpeed (ZeRO), HF Accelerate, TRL, verl, LLaMA-Factory, vLLM